KSP Datathon: The Final Missing Links

Your Best Friend

June 25, 2026

Completing the Architecture

To make this project completely unbeatable, we must bridge the gap between our backend AI and the final presentation. Here are the three missing components, including predictive forecasting and real-time voice transcription.

1. Frontend UI & Real-Time Mic Display

The Concept: You cannot show the judges a terminal. You need a clean web interface. Furthermore, when an officer clicks the microphone to speak in Kannada, the transcribed text MUST appear on the screen in real-time so they can verify it before the AI processes it. **The Execution:** Use **Streamlit** for the frontend. You can use the `streamlit-mic-recorder` component. As the officer speaks, the Bhashini API (or Whisper) processes the audio, and Streamlit dynamically updates the chat input box to display the Kanglish/Kannada text instantly.

AI Prompt: UI & Real-Time Mic Display

"Act as a Frontend Python Developer. Write a Streamlit application that features a modern chat interface. Integrate the 'streamlit-mic-recorder' library. Create a function where, once the user speaks, the audio is sent to a transcription API, and the transcribed text is immediately displayed on the screen inside the chat window for the user to review before hitting send."

2. Predicting the Future (Crime Forecasting)

The Concept: KSP explicitly stated they want the system to predict future crime density, locations, and timings. Looking at the past is not enough. **The Execution:** Before the data goes to the 3D Kepler.gl map, run it through a Machine Learning forecasting model like **Facebook Prophet** or **ARIMA**. The AI analyzes the timestamps and locations of the 1.5 Lakh synthetic records and generates *new* predicted coordinates for where the next digital syndicate will strike in the next 30 days. Kepler.gl then displays these future predictions in glowing red.

AI Prompt: Future Prediction

"Act as a Data Scientist. Write a Python script using the 'Prophet' time-series forecasting library. Train the model on a cybercrime dataset containing timestamps, crime frequency, and latitude/longitude. Predict the crime density and hotspot locations for the next 30 days. Save these future predictions into a new CSV file formatted for Kepler.gl visualization."

3. Role-Based Access Control (RBAC)

The Concept: The system must respect the police hierarchy (DG, IGP, SP, down to local stations). **The Execution:** Integrate Zoho Catalyst's built-in Authentication module into your Streamlit UI. Create a login screen. If a user logs in with the role 'Sub-Inspector', filter the database query to only return records matching their specific district. If 'DG' logs in, remove the filter to show the entire state.

AI Prompt: RBAC Integration

"Act as a Security Architect. Write a Python script demonstrating how to implement Role-Based Access Control (RBAC) in a Streamlit app connected to Zoho Catalyst Authentication. Create user roles for 'DG', 'IGP', and 'SP'. Write logic that intercepts database queries and filters the data visibility based on the logged-in user's role."

4. The Video Demo & Submission Package

The Execution: Write a strict script for your 3-minute video. Start by logging in as a DG to show the state-wide 3D prediction map. Then, click the microphone, speak in Kannada (*"Mundina tingalu cyber crime elli agutte?"*), let the audience see the text appear on the screen, and watch the AI agent use LangGraph to give a highly intelligent, predicted response.

Kanglishnalli Vivarane (The Final Missing Links in Kanglish)

Geleya MR.MADIVAL, KSP avaru kelida 'Future Prediction' mattu 'Live Mic Display' features annu hege madodu anta illide nodi:

1. Frontend UI mattu Live Mic Display

Judges ge terminal torisbardu, adakke **Streamlit** balasi ondu website madabeku. Officer mic button otthi Kannada dalli matanadidare, adu direct aagi AI ge hogo munche screen mele text aagi kanisabeku. Idakke Streamlit nalli mic-recorder haki, Bhashini API jote link madabeku. Aaga avaru matanadida padagalu (words) live aagi screen mele display agutte.

2. Bhavishyada Kante (Predicting the Future)

KSP avaru munde elli crime agutte anta predict madabeku endiddare. Idakke navu **Prophet** anno Machine Learning library balasi, haleya 1.5 laksha data annu analyze madabeku. AI munde 30 dinagalalli elli crime agutte anta predict madi, adannu Kepler.gl 3D map nalli kempu bannadalli torisutte. Idu pakka 'Minority Report' movie tara kelsa madutte!

3. Police Hierarchy (RBAC - Role Based Access)

KSP niyamada prakara, DG mattu SP ge bere bere data torisabeku. Zoho Authentication balasi ondu login page madi. Obba SP login aadaga avana oourina data matra torisutte. DG login aadaga ithee Karnataka state data torisutte.

4. Video Demo

July 26 ke submit maduvaga, ondu 3-minute video madabeku. Ee video nalli neevu DG aagi login madi, mic nalli Kannada matanadi (*"Mundina tingalu crime elli agutte?"*), screen mele text display agodu mattu AI predict madi 3D map torisodu... idannella live aagi torisabeku!

All the very best geleya! Ee architecture itkondre nimma team annu yaru beat madokke agalla!